

Image Analysis and Pattern Recognition

Final Project - 2026

UNO Vision

Detect, recognize, and infer the UNO game state from images



Problem Overview

Analyze images of UNO game and extract structured information about the game state.

For every image, your pipeline should predict:

- The **center card** on the table
- The **active player** (whose turn it is)
- The **set of cards** held by each of the four players

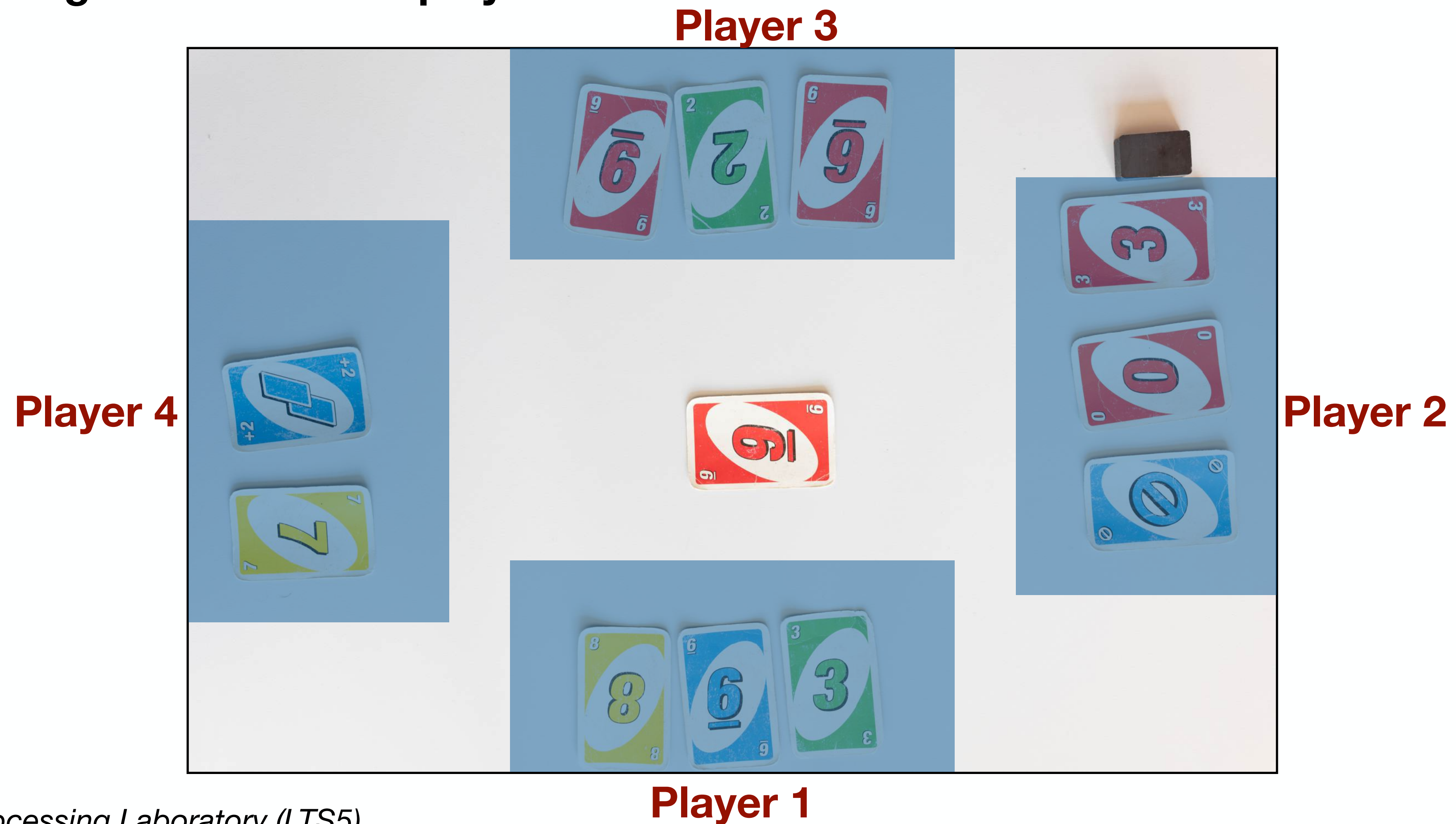


Example image

Game Setting

Player positions are **fixed** and must be respected.

⚠ **Not all images contain four players.**



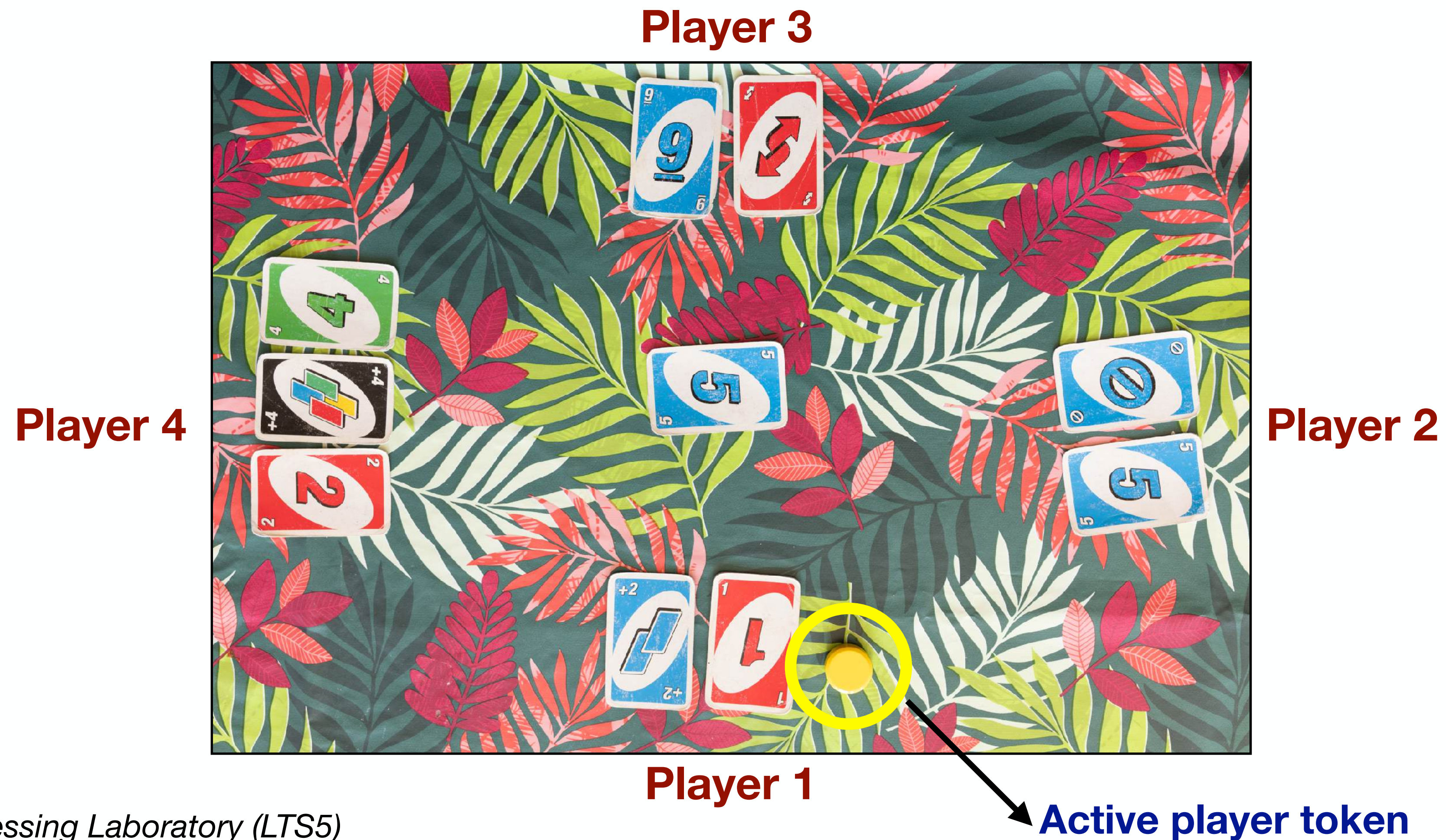
Game Setting

Active player is indicated by a token placed next to their cards.



Game Setting

Active player is indicated by a token placed next to their cards.



Dataset

- White background scenes
- Both non-overlapping and overlapping card layouts



Non-overlapping cards scene



Overlapping cards scene

Dataset

- Noisy background scenes
- Both non-overlapping and overlapping card layouts



Non-overlapping cards scene



Overlapping cards scene

Card Samples



Evaluation

For each image i , we evaluate:

- Center card prediction (10%)
- Active player prediction (10%)
- Card predictions per player (80%)

$$\text{CenterAcc}_i = 1[\hat{c}_i = c_i]$$

$$\text{ActiveAcc}_i = 1[\hat{a}_i = a_i]$$

$$F1_i = \frac{2TP}{2TP + FP + FN}$$

$$\text{Score} = 0.1 \cdot \text{CenterAcc} + 0.1 \cdot \text{ActiveAcc} + 0.8 \cdot F1$$

Kaggle Challenge

- Join the challenge with the invitation link in Moodle.
- **Read the detailed explanation on Kaggle for the dataset, submission format and rules!**

IAPR26: UNO Vision Challenge

Final project for EPFL IAPR course



[Overview](#) [Data](#) [Code](#) [Models](#) [Discussion](#) [Leaderboard](#) [Rules](#) [Team](#) [Submissions](#) [Settings](#)

Overview

In this challenge, you are given images of UNO game and asked to recover the game state from each image. For every image, your model should predict:

- The center card on the table
- The active player (whose turn it is)
- The set of cards held by each of the players

Start

2 minutes ago

Close

a month to go

Competition Host

Ekin Taskin



Prizes & Awards

Kudos [Edit](#)

Does not award Points or Medals

Participation

0 Entrants
0 Participants
0 Teams
0 Submissions

Tags

Custom Metric



Kaggle Challenge

- TAs baseline as reference —> outperforming the baseline in **private leaderboard** (+10 points)
- Compete against the baseline matching your approach (Classic or Deep Learning)

Leaderboard

[↓ Raw Data](#)[↻ Refresh](#)[Public](#) [Private](#)

This leaderboard is calculated with approximately 50% of the test data. The final results will be based on the other 50%, so the final standings may be different.

#	Team	Members	Score	Entries	Last	Join
	Deep Learning Baseline		0.64731	1	1h	
	Classic Baseline		0.57039	1	1h	

Important Rules

You are free to use classical image processing techniques, deep learning, or a combination of both.

⚠ The use of external datasets is strictly forbidden.

⚠ The test images should only be used for testing.

⚠ The use of pre-trained models is strictly forbidden.

⚠ The use of models with more than 12 million parameters is strictly forbidden.

• Violation of any rule: **-50 points on the final grade.**

Card Annotations

Cards are represented as strings, combining **the first letter of the color** and **value/action**.



b_0



r_4



y_9



g_8



y_skip



wild



g_reverse



r_draw_2



draw_4

Kaggle Submission Format

- The submission file must be a CSV with the following columns (see the sample_submission.csv):
image_id,center_card,active_player,player_1_cards,player_2_cards,player_3_cards,player_4_cards
- Multiple cards must be separated by a semicolon ; (e.g. r_5;b_skip;y_2)
- If a player has no cards, write: EMPTY
- active_player field is one of: p1, p2, p3, p4

```
image_id,center_card,active_player,player_1_cards,player_2_cards,player_3_cards,player_4_cards  
L1000777,r_7,p2,wild;b_4,r_reverse;y_3,r_7;b_draw_2,EMPTY  
L1000778,b_9,p4,EMPTY,b_1;y_7,b_3;g_2;y_reverse,b_5;r_2
```

Example submission

Kaggle Submission Format

Notes:

- ♣ Card order does not matter. (e.g. `b_5;g_reverse == g_reverse;b_5`)
- ♣ If a player holds the same card multiple times, it must be explicitly repeated.
- ♣ Players may be missing at arbitrary positions (EMPTY hands).
- ♣ Player positions are fixed.



Deliverables

-  Jupyter notebook report
 - Clear justification of design choices
 - Detailed technical descriptions of your solution
 - Strong quantitative and qualitative analysis
-  main.py (Executable Script)
 - Should produce the exact submission file uploaded to Kaggle
-  src/ Folder
 - All additional code and files used by main.py

Deadline: May 20th, 23:55

Final Presentation

 Date: Friday, May 22nd
 Location & Schedule: TBA

- Each group will deliver a **7-minute presentation** followed by an **8-minute Q&A** with the TAs.
 - Present your approach and key design decisions.
 - **Reason** and **interpret** your results:
 - Why is your pipeline working / not working?
 - What would you improve next?
- ★ *We are interested not only in competition results, but in your understanding and reasoning.*
- 🧐 *Be prepared to answer questions about your approach and ensure that you understand how your algorithm works.*

Final Presentation

 Date: Friday, May 22nd
 Location & Schedule: TBA

- If you follow a classical pipeline, you should clearly explain and demonstrate:
 - Segmentation
 - Object Description
 - Classification
- If you follow a deep learning approach, you should:
 - Justify your design choices (e.g. architecture, loss, data processing)
 - Provide insights into what your model has learned (e.g. visualization or interpretability techniques).
Demonstrate your model learns object locations and features.
- For all approaches, you should include **failure case analyses** and discuss possible reasons for errors.

Demonstrate these analyses in your Jupyter notebook report as well.

QUESTIONS ?

For the project-related questions, use the ed project forum in Moodle.

